

Armando Santos

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github.com/bolt12 · hackage.haskell.org/user/bolt12

Haskell engineer, 5 years consulting at Well-Typed. Published a functional pearl at the Haskell Symposium and maintain several packages on Hackage. Recent side interests consist of maintaining a small NixOS server fleet and the local LLM stack on it; also cooking, working out and learning Japanese.

SKILLS

Programming	Haskell, C, Nix, Python, JavaScript
Distributed systems	P2P networking, concurrency, property-based testing
Infrastructure	NixOS, ZFS, Prometheus and Grafana, Terraform
Local LLM	llama.cpp, vision-language pipelines
Formal methods	Agda, Alloy, Coq, Z3
Languages	Portuguese (native), English (fluent), Japanese (JLPT N4)

EXPERIENCE

Well-Typed LLP, Haskell Consultant

Remote · March 2021 to present

hs-bindgen (June 2025 to present)

Well-Typed's tool for generating Haskell bindings from C headers. I work across most of it:

- Extended C language coverage and improved the generated code.
- Built the Doxygen to Haddock pipeline so C header documentation reaches the generated Haskell, and split the parser out as *doxygen-parser*.
- Fixed and expanded Template Haskell support and cross-compilation, and wrote parts of the manual.
- Built *binding-combinators*, a high-level bindings library, most recently.

Cardano networking, on assignment to IOG (March 2021 to April 2025)

Four years on the P2P layer of *ouroboros-network*.

- Designed and implemented the Peer Sharing and Bootstrap Peers mini-protocols on *typed-protocols*.
- Built the internal mainnet and testnet deployments with Terraform and Nixops.
- Made the network layer polymorphic over its parameters and protocols, so it no longer depended on Cardano specifics.
- Spoke at OPODIS 2022 and wrote the engineering post on the Dynamic P2P design.

Training

Prepared and gave the online lectures for Well-Typed's introductory Haskell course, and reviewed student assignments.

INESC TEC, Research Assistant

Braga, Portugal · June to December 2020

Built a Haskell library for expressing probabilistic problems in the linear algebra of programming style, using Selective functors as the underlying abstraction. It became *laop* on Hackage.

PUBLICATIONS AND TALKS

Type your matrices for great good: a Haskell library of typed matrices and applications, (functional pearl).

Santos and Oliveira, ACM SIGPLAN Symposium on Haskell, 2020. doi.org/10.1145/3406088.3409019

Verifying and testing timeliness constraints with io-sim, Well-Typed blog, 2025.

Unveiling Cardano's Dynamic P2P: a leap forward in decentralization, IOG Engineering blog, 2023.

OPODIS 2022, industry talk on the design decisions behind the IOG networking team's P2P work.

PROJECTS

nixos github.com/bolt12/nixos

My personal NixOS configuration and homelab. About 25 services run on the server, all declared in Nix.

pet-report github.com/bolt12/pet-report

A daily journal of what my cat and dog get up to, built on *Frigate* and a local vision model. Haskell backend, Svelte frontend, ships as its own flake and NixOS module. It solves a real problem for me: knowing how my pets *actually* did when I am out of the house for long periods, and whether either of them needs attention.

OPEN SOURCE

- Hackage: *laop*, *f-algebra-gen*, *shannon-fano*, *data-combinator-gen*, *doxygen-parser*.
- Contributed to *agda-stdlib* (base-2 logarithms and their properties) and *agda-categories* (star-autonomous categories).
- Contributions to *io-sim* (timeouts API refactor) and *algebraic-graphs* (undirected graph support).

EDUCATION

MSc Computer Engineering, University of Minho, 2015 to 2021. Thesis: Selective Applicative Functors and Probabilistic Programming.

Track owner at *Haskell Love 2020*. Volunteer at *FM'19* in Porto.

Federated sprinter for SC Braga.